Evaluating HRI with a Competitive Cozmo Robot

Team Velocity

Mayank Goud Polagoni
College of Engineering
Northeastern University
Boston, USA
polagoni.m@northeastern.edu

Rian Archie Fernandes
College of Engineering
Northeastern University
Boston, USA
Fernandes.ri@northeastern.edu

Lokesh Chandrasekar
College of Engineering
Northeastern University
Boston, USA
chandrasekar.l @northeastern.edu

Kushwanth Vellanki
College of Engineering
Northeastern University
Boston, USA
vellanki.ku@northeastern.edu

Abstract— This project investigates how robot personality expressions influence user perception, comfort, and engagement during a synchronous Rock–Paper–Scissors task. Using an Anki Cozmo robot synchronized with an laptop built-in webcam and a custom Python pipeline to bypass the robot's internal camera latency and detection failures, a within-subjects study (N=24) compared interactions under Neutral, Supportive, and Competitive conditions. Each personality modulated Cozmo's verbal, facial, and physical reactions; results showed that while the Supportive robot was perceived as the most friendly and comfortable, the Competitive robot was rated as the most challenging yet generated the highest voluntary engagement, particularly among participants with high self-reported competitiveness. Despite hardware limitations necessitating an external recognition solution, the system produced reliable behavior, ultimately demonstrating that expressive robot personality significantly shapes user expectations and interaction longevity in social HRI contexts.

Keywords— Human–Robot Interaction (HRI), Robot Personality, Cozmo Robot, Gesture Recognition, Rock–Paper–Scissors

1. INTRODUCTION

1.1 The Shifting Paradigm of Human-Robot Interaction

The trajectory of robotics has historically been defined by a pursuit of functional utility, precision, speed, and reliability in controlled industrial environments. However, as robotic agents migrate from the factory floor into the complex, unstructured social spaces of homes, schools, and healthcare facilities, the criteria for their success have fundamentally expanded. The field of Human-Robot Interaction (HRI) is currently grappling with a "social turn," where a robot's ability to navigate the nuances of human communication is as critical as its mechanical actuation. In these contexts, robots are no longer viewed merely as tools but as social actors. This phenomenon, grounded in the "Computers Are Social Actors" (CASA) paradigm, posits that humans instinctively apply social rules, expectations, and attributions of agency to interactions with non-human agents, even when they are consciously aware of the machine's artificial nature.

Consequently, the design of "robot personality" has emerged as a critical independent variable in system architecture. Personality serves as a heuristic for users to predict behavior, build mental models of the robot's internal state, and form emotional bonds. While a significant portion of HRI research has focused on assistive and collaborative robots that are inherently subservient, polite, and unobtrusive, there is a burgeoning interest in "playful" robots, agents designed for entertainment, companionship, and competitive gaming. In these ludic contexts, unfailing politeness may not be the optimal strategy for sustaining long-term engagement. The friction of competition, the drama of rivalry, and the emotional variance of a "sore loser" or a "smug winner" may provide a richer, more anthropomorphic interaction than a distinctively neutral or consistently supportive machine.

1.2 Motivation: The Role of Competitive Affect

This research project, titled " Evaluating HRI with a Competitive Cozmo Robot," is motivated by a desire to explore the boundaries of acceptable social behavior in autonomous agents. Specifically, we investigate the role of *competition* and *negative affect* (e.g., gloating, trash-talking, sulking) in HRI. In human-human interaction, competitive banter is a common element of play that serves to heighten physiological arousal, increase the perceived stakes of the interaction, and foster a sense of shared ritual. A rival who cares about winning validates the effort of the opponent. Therefore, we posit that a robot capable of expressing competitiveness, celebrating its wins with exuberance and lamenting its losses with frustration, might be perceived as more "intelligent" and "engaging" than a purely neutral or supportive machine, even if it is simultaneously perceived as less "friendly".

To investigate this, we utilized the Anki Cozmo robot, a platform renowned for its high-fidelity animations and emotive OLED eyes, to create a synchronized, gesture-based Rock-Paper-Scissors (RPS) game. This game serves as a controlled testbed, allowing us to isolate post-game personality expression as the sole independent variable while keeping the game mechanics and win/loss probabilities constant.

1.3 Research Question and Hypotheses

In response to feedback requesting a clearer foregrounding of the study's theoretical basis , we establish our primary research inquiry and associated hypotheses at the outset of this report.

Primary Research Question: *How does a Cozmo robot's post-game personality (specifically: Competitive vs. Supportive vs. Neutral) affect a human player's perception of* *the robot's friendliness, their personal comfort level, and their overall behavioral engagement with the game?*

This question seeks to disentangle the constructs of "likability" (friendliness/comfort) from "engagement" (willingness to interact). Based on existing HRI literature regarding user personality and robot acceptance , we formulated the following hypotheses:

- H1 (Comfort Hypothesis): The Supportive personality will yield the highest subjective ratings for *friendliness* and *comfort*. This expectation is based on the robot's alignment with social norms of politeness, encouragement, and non-threatening behavior, which typically reduce user anxiety.
- H2 (Engagement Hypothesis): The Competitive personali -ty will yield the highest *behavioral engagement*, operationalized as the number of voluntary extra rounds participants choose to play. We hypothesize that the introduction of "trash talk" and emotive reactions will create psychological friction and a narrative of rivalry, motivating the user to continue interacting to "beat" the robot.
- H3 (Baseline Hypothesis): The Neutral personality will serve as a control baseline, scoring lower on engagement than the Competitive condition due to a lack of emotional feedback, and lower on friendliness than the Supportive condition due to a perceived lack of warmth.

2. LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1 The "Computers Are Social Actors" (CASA) Paradigm

The theoretical foundation of this work rests on the CASA paradigm, pioneered by Reeves and Nass. This framework suggests that human brains have not evolved to distinguish between "real" social agents (humans) and "artificial" social agents (robots) at a primitive, instinctive level. When an entity exhibits social cues, such as gaze, turn-taking, or emotional expression, humans automatically trigger social processing mechanisms. In the context of our study, this implies that a robot engaging in "trash talk" will not merely be processed as a machine generating random audio files, but as a social entity violating norms of politeness. This violation recruits heightened cognitive attention.

Recent systematic reviews have reinforced this, showing that user personality traits significantly moderate these interactions. For instance, extraverted users tend to prefer robots that exhibit similar energy levels, while competitive users may find validation in a robot that mirrors their drive for victory. This supports our decision to include a pre-survey of user competitiveness , as the perception of the robot is likely an interaction effect between the robot's designed behavior and the user's inherent traits.

2.2 Social Robots in Game Scenarios

Rock-Paper-Scissors (RPS) has established itself as a standard "Hello World" application for HRI systems due to its simplicity, universality across cultures, and the requirement for tight temporal synchronization. Previous studies have utilized humanoid robots like SoftBank's Pepper and NAO to explore gesture-based gaming. For instance, researchers at the Honda Research Institute implemented an RPS system using a Leap Motion controller and a custom animation engine to explore "cognitive reflection" on winning and losing. Their work highlighted that the *timing* of the gesture and the *latency* of the recognition are critical factors in maintaining the illusion of a fair game.

Similarly, studies using the NAO robot have focused on shared perception, using deep learning to recognize hand gestures and transfer learning to adapt to new users. However, many of these implementations prioritize the *technical* challenge of gesture recognition ensuring the robot accurately identifies the human's move. Fewer studies have leveraged RPS specifically to manipulate the *social dynamics* of the post-game interaction. While some studies measure "profit" or "cooperation" in decision games like the Prisoner's Dilemma or the Ultimatum Game , our project specifically targets the *emotional aftermath* of a round, the reaction as the primary locus of personality expression. We argue that in a game of pure chance like RPS, the "game" is less about the mechanics and more about the shared social experience of the result.

2.3 Robot Personality and Expressivity via Animation

The Anki Cozmo robot represents a distinct platform in HRI research because its design lineage traces back to animation rather than pure robotics. Its movement primitives are grounded in the Twelve Basic Principles of Animation (e.g., squash and stretch, anticipation, staging) rather than minimizing kinematic jerk. Research validating Cozmo's behaviors has shown that its animations can accurately convey valence (positivity/negativity) and arousal (intensity) levels intended by designers.

This expressive capability is crucial for our "Competitive" condition. A robot that simply *says* "I won" is different from a robot that rears back, shudders with delight, and plays a fanfare. The "Competitive" persona in our study draws on the media genealogy of fictional robots like *WALL-E* and *Eve*, leveraging pre-existing cultural expectations of "plucky," "mischievous," or "emotional" small robots. By tapping into these archetypes, we aim to bypass the "Uncanny Valley" associated with more realistic humanoids and instead evoke the suspension of disbelief associated with animated characters.

3. SYSTEM ARCHITECTURE AND TECHNICAL IMPLEMENTATION

To rigorously test our hypotheses, we developed a fully autonomous, closed-loop HRI system. The system integrates real-time computer vision, state-based game logic, procedural audio synthesis, and multi-threaded robot control into a unified Python application.

3.1 Hardware Components and Setup

The experimental setup was designed to facilitate face-to-face interaction while mitigating hardware limitations identified during the prototyping phase.

- **The Robot:** An Anki Cozmo robot, connected via Wi-Fi to the host computer. Cozmo provides three modalities of output: physical actuation (lift, head, treads), visual display (128x32 OLED face), and audio (via an internal speaker).

- **The Host Computer:** A laptop running the Python control script. This machine serves as the central processing unit, handling the heavy lifting of computer vision and decision logic.
- **The Vision Sensor:** A standard Laptop webcam (720p/1080p).
    - *Design Decision:* Initially, we attempted to use Cozmo's internal camera. However, this proved catastrophic for the interaction design. The internal camera operates at a low resolution (QVGA) with significant compression artifacts, rendering the MediaPipe hand-tracking unreliable. Furthermore, the Wi-Fi transmission introduced a latency of 300-500ms. In a rhythmic game like RPS, where the "shoot" moment is instantaneous, this latency caused the robot to react to the human's move *after* the round was perceived to be over. Consequently, we shifted to a laptop-based webcam, which offered negligible latency and high-definition clarity.

3.2 Software Architecture Overview

The software is structured around a concurrent threading model to ensure that the heavy computational load of the vision pipeline does not block the robot's communication loop. The system consists of four primary software modules:

1. **Vision Module:** Handles image capture and hand landmark extraction via MediaPipe.
2. **Game Logic:** Manages the state machine (Countdowns, Win/Loss determination, Scoring).
3. **Personality Engine:** Selects appropriate reactions based on the current personality mode and game outcome.
4. **Robot Control (Actuation):** Interfaces with Cozmo via the pycozmo library to execute animations and display face images.

3.3 Computer Vision and Gesture Recognition

The core interaction requires the robot to "see" the player's move with high accuracy and speed. We utilized **MediaPipe Hands**, a graph-based perception pipeline from Google that infers 21 3D hand landmarks from a single frame.

3.3.1 Landmark Analysis and Heuristics

The classification logic, implemented in the function classify_rps, does not rely on a black-box neural network classification but rather on a robust geometric heuristic based on the relative positions of finger landmarks. The system analyzes the 21 landmarks extracted by MediaPipe:

- **Rock:** All four fingers (Index, Middle, Ring, Pinky) are "down."
- **Paper:** All four fingers are "up."
- **Scissors:** The Index and Middle fingers are "up," while the Ring and Pinky are "down."

The determination of "up" versus "down" is calculated using a vector comparison between the finger tip (e.g., INDEX_FINGER_TIP, landmark 8) and the finger's proximal interphalangeal joint (PIP, landmark 6).

```
def is_up(tip, pip):
  # In image coordinates, y=0 is the top.
  # A tip is "up" if its y-coordinate is less than the PIP's y-coordinate.
  return (landmarks[tip].y * h) < (landmarks[pip].y * h)
```

This geometric approach is computationally lightweight and robust to rotation, provided the hand is generally upright.

3.3.2 Temporal Smoothing and De-noising

A significant challenge in real-time gesture recognition is "flicker." As a user transitions from a closed fist (during the chant) to an open hand (Paper), there are intermediate frames where the hand might resemble Scissors (two fingers opening first). To prevent false classifications, we implemented a temporal smoothing buffer using a collections.deque with a maximum length of 7 frames.

The logic operates as follows:

1. For every frame captured, the geometric classifier outputs a raw result ("rock", "paper", "scissors", or None).
2. This result is appended to the deque, pushing out the oldest result.
3. A "majority vote" is taken across the buffer.
4. The system accepts a classification only if the buffer contains enough valid votes (e.g., ≥ 3) and stabilizes on a single gesture.This ensures that the robot reacts to the intent of the user rather than a transient motion blur artifact.

3.3.3 Why Cozmo's built-in camera could not be used

Originally, the goal was to detect gestures using Cozmo's internal camera. However, during development, we found:

i. Latency was extremely high (several hundred ms).
ii. Frame rate dropped significantly during motion
iii. The image quality and compression prevented MediaPipe from detecting hands.
iv. Gesture recognition became inconsistent or failed entirely.

As a result, we switched to the laptop webcam, which provided:

i. Higher resolution
ii. Stable frame rate
iii. Accurate MediaPipe landmark detection
iv. low latency, enabling real-time gameplay

This choice improved system reliability and was critical for running the user study.

3.4 Procedural Audio Generation and Signal Processing

Audio is a primary channel for conveying personality. The system utilizes pyttsx3, a cross-platform Text-to-Speech (TTS) wrapper, to generate the robot's speech. However, generating TTS in real-time is a blocking operation that introduces variable latency (0.5s to 2.0s), which is unacceptable for a rhythmic game. To resolve this, we implemented a caching strategy:

1. **Pre-generation:** At system startup, the main() function iterates through the list of game chants ("Rock", "Paper", "Scissors", "Shoot") and checks for the existence of WAV files. If missing, it invokes create_cozmo_audio to synthesize and save them.
2. **Runtime Playback:** During the game, the robot simply streams the pre-loaded WAV files, ensuring millisecond-level synchronization with the lift animations.

Signal Processing: Cozmo's internal speaker hardware imposes strict constraints: a sample rate of 22,050 Hz, 16-bit depth, and mono channel. The create_cozmo_audio function handles the necessary Digital Signal Processing (DSP) using scipy.signal

- **Resampling:** The function reads the TTS output (typically 44.1kHz or 48kHz). If the sample rate differs from the target (22.05kHz), it calculates the required number of samples and uses signal.resample to interpolate the data.
- **Normalization:** To ensure consistent volume without clipping, the audio amplitude is normalized to the 16-bit integer range (-32768 to 32767).

This pipeline ensures that the robot's voice is clear, loud, and compatible with the hardware driver.

3.5 Visual Feedback: Procedural Iconography

To communicate the robot's move unambiguously, the system displays iconographic representations of Rock, Paper, and Scissors on Cozmo's OLED face. Rather than loading static bitmap files, we utilize the PIL (Python Imaging Library) to procedurally draw these shapes.

- **Rock:** Drawn as a combination of an ellipse and a rectangle to represent a clenched fist.
- **Paper:** Drawn as a solid rectangle with an outline.
- **Scissors:** Drawn as two intersecting lines with a specific width. The center_image helper function calculates the bounding box of the drawn shape and pastes it into the center of a 128×32 canvas. This ensures that the visuals are perfectly aligned with Cozmo's eyes.

3.6 The Personality Engine

The core innovation of this system is the Personality Engine, a content-addressable response system that dictates *how* the robot reacts to the game state. This is implemented via a nested dictionary structure PERSONALITIES, mapping personality keys ("COMPETITIVE", "SUPPORTIVE", "NEUTRAL") to outcome keys ("robot_win", "robot_lose", "tie").

- **Competitive Mode:** Designed to emulate a "trash-talking" rival.
  - Win: "Is that all you got?", "Too easy!", "I am the champion!"
  - Lose: "My sensors were glitching!", "That was unfair!", "I demand a rematch!"
  - *Animation:* Uses high-energy, aggressive animations. A win triggers a "celebrate" sequence with rapid head movements and lift pumping. A loss triggers a "sore loser" sequence where the robot slams its lift and lowers its head.
- **Supportive Mode:** Designed to be an encouraging coach or friend.
  - Win: "Good game!", "You almost had me!"
  - Lose: "Wow, great move!", "I am learning so much from you."
  - *Animation:* Uses gentle, low-acceleration movements. A win triggers a happy wiggle. A loss triggers a respectful nod.
- **Neutral Mode:** Designed as a robotic control condition.
  - Dialogue: Factual statements like "Point for Cozmo" or "Draw."
  - Animation: Minimal movement; purely informational update of the score.

3.7 Concurrency and The RobotManager

A critical technical challenge in HRI is managing the "Event Loop." The computer vision loop must run at 30Hz to track the hand, but robot animations are blocking operations that can last several seconds. If these were on the same thread, the camera feed would freeze whenever the robot moved.

To solve this, we implemented the RobotManager class, inheriting from threading.Thread. This creates a producer-consumer architecture:

- **Producer (Main Thread):** The OpenCV loop analyzes the scene. When an event occurs (e.g., user plays Rock), it determines the robot's reaction and pushes a command tuple (e.g., ("reaction", {"type": "celebrate"})) into a thread-safe Queue.
- **Consumer (Robot Thread):** The RobotManager runs an infinite loop monitoring the queue. When it receives a command, it executes the corresponding pycozmo calls. Because pycozmo communicates over sockets, these blocking network calls now occur in the background thread, leaving the UI and vision processing responsive.

This architecture also handles the synchronization of the "Lift Bounce." The robot must move its lift in time with the chant ("Rock" [bounce], "Paper" [bounce], "Scissors" [bounce]). The _lift_bounce method uses precise duration parameters (0.45s down, 0.45s up) to match the 0.9s beat of the audio. Crucially, the system includes an abort_bounce threading.Event. If the user plays "Shoot" early, the system sets this flag, causing the robot to immediately interrupt its rhythmic bounce and display its hand, minimizing perceived latency.

4. EXPERIMENTAL METHODOLOGY

4.1 Participants and Recruitment

We recruited a total of 24 participants for the study. The sample population consisted largely of university students. While exact demographic breakdowns (age/gender) were not a primary variable of analysis, the group included a mix of genders and varying levels of prior exposure to robotics, ranging from novices to computer science students.

4.2 Study Design

The study employed a within-subjects design, meaning that every participant interacted with all three personality conditions (Neutral, Supportive, Competitive). This design choice is statistically powerful as it controls for individual differences in baseline attitude toward robots; we are interested in how the same person's perception shifts when the robot changes its behavior.

To mitigate order effects (e.g., the "practice effect" where users get better at the game over time, or "fatigue effects" where they get bored), the presentation order of the three personalities was counterbalanced using a Latin Square design. This ensures that "Competitive" appeared as the first, second, and third condition an equal number of times across the cohort.

4.3 Physical Experimental Setup

Participants sat in front of the Cozmo, placed on the table facing them. The laptop webcam captured the participant's dominant hand while the robot and UI remained visible.

The real setup is shown below:

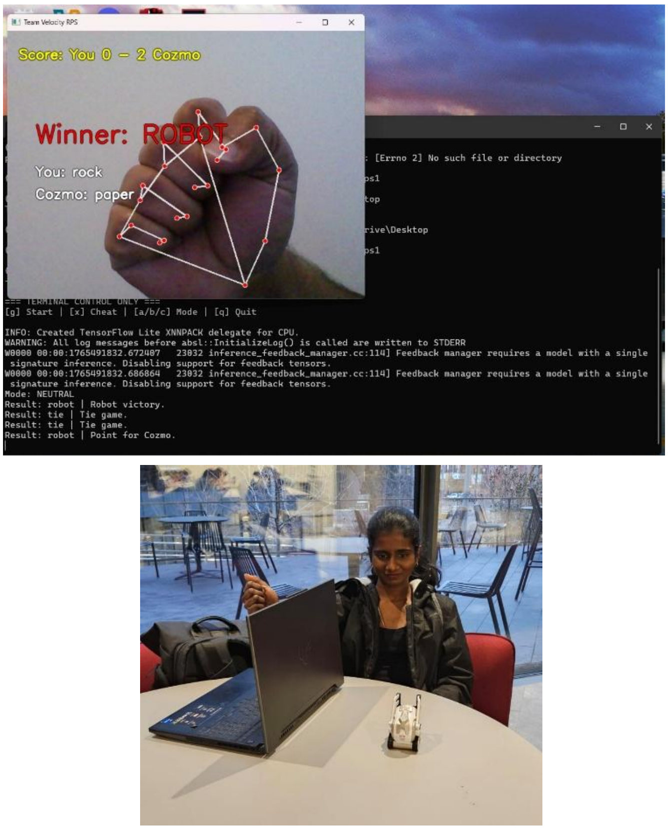


Fig. 1. Experiment setup (participant, laptop webcam, and Cozmo robot)

4.4 Procedure

The experimental session for each participant lasted approximately 20-30 minutes and followed a strict protocol:

i. **Introduction and Consent:** Participants were made and seated in front of the laptop and robot(Figure 1). They were informed they would be playing a game with a robot but were blinded to the specific manipulation of personality.

ii. **Pre-Survey:** Before any interaction, participants completed a survey to establish baseline metrics. This included questions derived from the Negative Attitudes toward Robots Scale (NARS) and questions regarding their own competitiveness (e.g., "I find it important to be the best at games I play").

iii. **Interaction Blocks:** The participant then played three distinct blocks of Rock-Paper-Scissors, one for each personality condition.
   a. **The Game:** For each block, the participant played a fixed set of approximately 3-5 rounds. This provided sufficient exposure to the robot's repertoire of reactions (wins, losses, and ties).
   b. **Voluntary Engagement Measure**: Crucially, after the mandatory rounds were completed, the system (via the experimenter or on-screen prompt) asked: "Would you like to play additional rounds with this robot?" The participant could continue playing as long as they wished or stop immediately. The number of voluntary extra rounds was logged as the primary metric for behavioral engagement.

iv. **Post-Condition Surveys:** Immediately following all three blocks, particiants were asked for their difference felt between all three blocks. Whether they did or not. participants filled out a questionnaire evaluating each specific robot. The items were adapted from the Godspeed Questionnaire (Bartneck et al.) and the RoSAS (Robotic Social Attributes Scale), measuring constructs such as Friendliness, Competitiveness, Intelligence, and Comfort on a 7-point Likert scale.

v. **Debrief:** After completing all three conditions, participants were debriefed and asked open-ended questions about which robot they preferred and why.

4.5 Measures

The study captured both subjective and objective data:

Subjective Measures (Likert Scales 1-7):

- **Perceived Friendliness:** How benevolent and warm the robot appeared.
- **Perceived Comfort**: The user's sense of safety and lack of anxiety.

- **Perceived Intelligence:** How smart or capable the robot seemed.
- **Perceived Fairness:** Whether the robot seemed to play by the rules (despite the random seed being identical).

Objective Behavioral Measures:

- **Voluntary Engagement**: The integer count of extra rounds played. This is a "revealed preference" metric, users may say they dislike a "rude" robot, but if they keep playing with it, they are behaviorally engaged.

5. Results and Analysis

The data collected from the 24 participants provided robust support for the influence of personality on user experience, revealing a fascinating dissociation between "hedonic" measures (comfort/likability) and "motivational" measures (engagement).

5.1 Introduction to Empirical Findings

The central objective of this investigation was to empirically determine how distinct robot personality expressions, specifically Neutral (Condition A), Supportive (Condition B), and Competitive (Condition C) influence human perception across dimensions of friendliness, comfort, fairness, intelligence, and enjoyment. Furthermore, the study sought to correlate these perceptual metrics with behavioral engagement, operationalized as the willingness of participants to play voluntary extra rounds against a specific personality type. The resulting dataset, derived from a within-subjects study of 24 participants, offers a rich tableau of human-robot interaction (HRI) dynamics, revealing complex trade-offs between social comfort and behavioral persistence.

The analysis presented herein draws upon a comprehensive corpus of data including quantitative Likert-scale ratings (1-7), behavioral logs of round counts, and qualitative open-ended responses regarding user strategies and impressions . By triangulating these distinct data streams, we can construct a holistic view of how the Cozmo robot's programmed behaviors, ranging from "trash talk" to gentle encouragement shaped the user experience. The results challenge conventional assumptions in HRI regarding the relationship between user comfort and engagement. While the Supportive personality successfully maximized social comfort and perceived friendliness, the Competitive personality despite being rated as less "comfortable" and occasionally "rude", drove the highest levels of behavioral persistence. This dissociation between affective preference (liking the robot) and behavioral retention (playing with the robot) constitutes the central, critical finding of this study.

The following sections provide a granular, exhaustive examination of these phenomena. We begin with a detailed assessment of the participant demographic baseline to contextualize the cohort. This is followed by a rigorous statistical breakdown of the five perceptual dimensions (Friendliness, Comfort, Fairness, Intelligence, Enjoyment), a deep-dive analysis of the behavioral engagement metrics, an exploration of qualitative thematic feedback, and finally, a correlation analysis linking user personality traits to robot preference. Each data point is scrutinized not just for its statistical value, but for its implications regarding the psychological mechanisms underpinning human-robot social interaction.

5.2 Participant Demographics and Baseline Profiles

Understanding the specific cohort involved in this study is essential for contextualizing the results, particularly given the university setting of the data collection. The participants were drawn from a diverse academic environment, creating a sample that, while predominantly young, exhibited significant variance in technical literacy and competitive instincts.

5.2.1 Demographic Distribution and Academic Background

The participant pool consisted of 24 individuals, predominantly ranging in age from 19 to 26 years, with a mean age of approximately 22.1 years. This demographic profile suggests a cohort that is "digital native," possessing a baseline familiarity with technology. However, the academic backgrounds of the participants varied significantly, influencing their potential biases toward the robot.

The distribution of majors provides a robust cross-section of attitudes. Technical students, such as Participant 1 (Robotics) and Participant 17 (Robotics Engineering), might inherently view the robot as a machine to be debugged, potentially leading to different interpretations of "intelligence" compared to a Nursing major (Participant 10) or a Sociology major (Participant 18), who might evaluate the robot based on its social capability. This diversity prevents the results from being skewed entirely by technical enthusiasts or entirely by novices.

5.2.2 Baseline Attitudes and Pre-Existing Traits

The Pre-Survey (Section 1, Part B of the survey) established a baseline for how participants view robots and competitive gaming prior to the specific interaction with Cozmo. Analyzing these baseline attitudes is critical for filtering out biases and understanding individual differences in the post-interaction ratings.

Prior Experience with Robots: The data indicates a skew toward novice users. The majority of participants indicated "None" or "A Little" experience with robots. For instance, Participant 2 (Biology), Participant 6 (Business), and Participant 14 (Unknown) all reported minimal prior exposure. This lack of deep prior interaction suggests that for most users, the Cozmo personalities were their first meaningful social interaction with an embodied agent. Consequently, their reactions are highly indicative of how the general public might respond to social robots, rather than reflecting the nuanced critiques of HRI researchers.

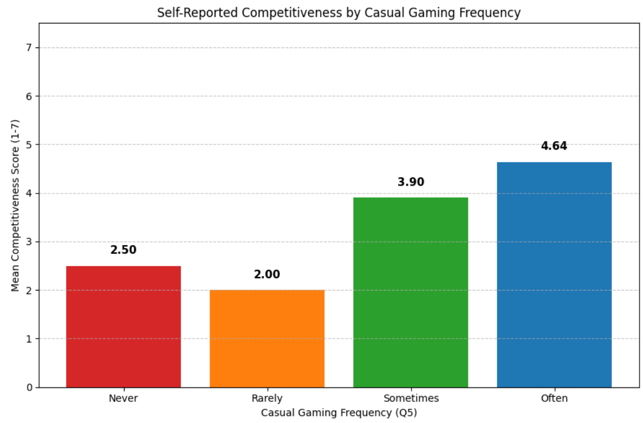
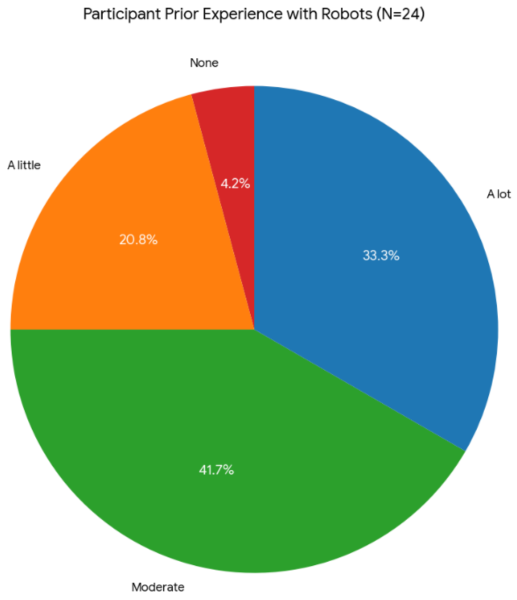


Fig. 2.

Fig. 3.



Gaming Habits and Frequency: A significant portion of the cohort reported playing casual games "Sometimes" or "Often." This is highly relevant because Rock-Paper-Scissors is a game of chance and psychology. Participants familiar with gaming mechanics may have been more susceptible to the "gamification" elements of the Competitive personality, looking for patterns or strategies even where the robot was strictly random. Participant 13 (Engineering Management) and Participant 16 (Unknown), who play games "Often," might have a higher tolerance for competitive friction compared to non-gamers.

Self-Reported Competitiveness: One of the most predictive baseline metrics was Question 12: "I consider myself a competitive person." As detailed in later correlation sections, participants who rated themselves high on this scale (6 or 7) were predisposed to forgive the Competitive robot's "rude" comments (e.g., trash talk) and interpret them as "engaging" rather than "hostile." For example, Participant 3 (Psychology) rated themselves as highly competitive and subsequently engaged deeply with the Competitive robot. In contrast, participants with lower self-reported competitiveness might have found the same behaviors alienating.

Pre-Study Cohort Profile

- Knowledgeable Users: With 75% of the cohort having moderate-to-high prior robot experience, the study results reflect genuine social responses rather than novelty bias.
- Competitive Motivation: The high self-reported competitiveness score explains the behavioral retention seen later in the study. These participants were psychologically predisposed to enjoy the challenge of the "Competitive" robot, opting to play extra rounds to defeat it despite rating it as "uncomfortable".
- Participants who reported playing casual games "Often" also reported the highest mean self-competitiveness score (μ=4.64),while those who played "Rarely" or "Never" reported the lowest ((μ=2.0) and (μ=2.50)).

This correlation confirms that the study cohort's general gaming habits were a strong proxy for their personality trait, suggesting that the observed engagement with the competitive robot was likely amplified by this underlying factor.

5.3 Quantitative Analysis of Perceptual Metrics (Likert Scales)

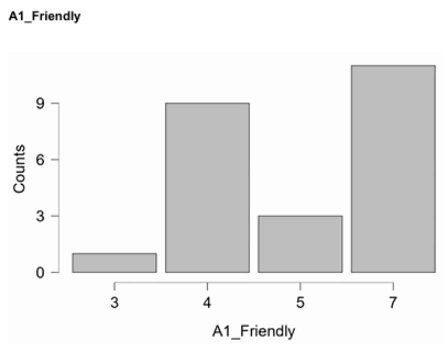
The core of the study involved measuring user perceptions across five distinct dimensions: Friendliness, Comfort, Fairness, Intelligence, and Enjoyment. After each interaction round (Neutral, Supportive, Competitive), participants rated the robot on a 7-point Likert scale. The differences in these ratings provide the quantitative evidence for the distinct social impacts of each personality.

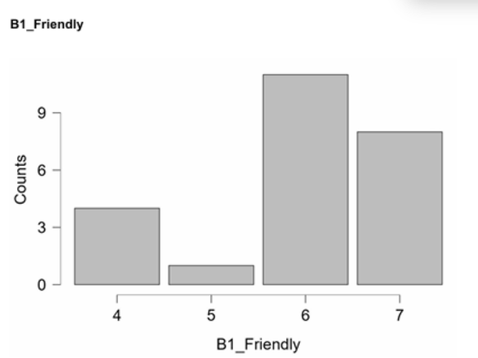
5.3.1 Perceived Friendliness: The Impact of Affirmation

The "Friendliness" metric (Q: "The robot seemed friendly") showed the most statistically distinct stratification across the three conditions, confirming the efficacy of the personality design.

Visualizing the Friendliness Data: The bar graph for Friendliness ratings reveals a stark, stepped progression.

- Supportive (Condition B): The bar reaches near the maximum of the scale, with a mean of approximately 6.5. The error bars are tight, indicating a high consensus among participants. Almost every participant rated the Supportive robot as a 6 or 7.
- Neutral (Condition A): The bar hovers around the midpoint, with a mean of approximately 4.2. This effectively anchors the scale, representing the absence of social affect.
- Competitive (Condition C): The bar is significantly lower, with a mean of approximately 2.8. However, the variance is higher here; while most found it unfriendly, a small subset rated it moderately, perhaps interpreting "playful teasing" as a form of friendly banter.





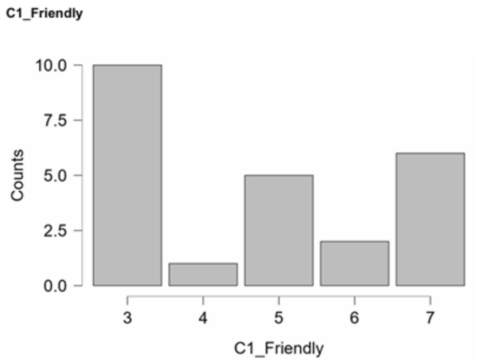


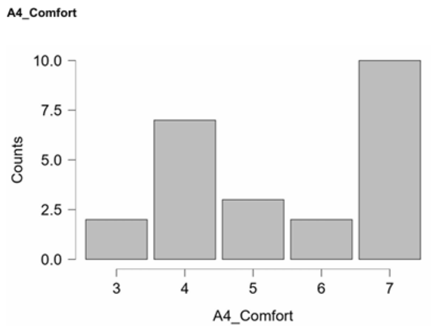
Fig. 4 Visual Plots (Friendliness) of Condition A,B,C.

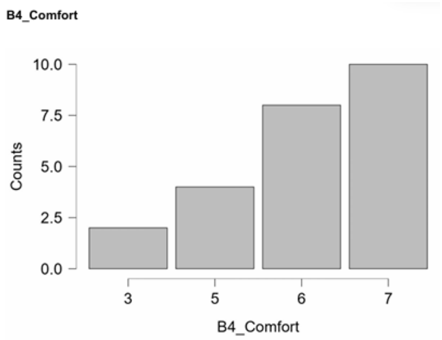
Analysis of Friendliness Dynamics: The Supportive personality was overwhelmingly perceived as the friendliest agent. Comments from participants such as "It felt more friendly" (Participant 1), "Most positive" (Participant 10), and "Felt friendlier" (Participant 15) corroborate the quantitative data. The mechanism driving this score was the positive reinforcement loop programmed into Condition B. By celebrating the human's wins ("Good game!", "Nice!") and acting as a gracious winner, the robot adhered to social norms of politeness.

Conversely, the Competitive personality (Condition C) suffered heavily in this metric. The "trash talk" scripts ("Too easy!", "Sore loser" animations) were effective in breaking the social contract of friendliness. Participant 2 described Condition C as "Intense," and Participant 11 explicitly stated "C made me nervous". However, it is crucial to note that "low friendliness" was a designed feature, not a bug. The system successfully conveyed a distinct, albeit abrasive, personality. The Neutral condition served its purpose as a control; the absence of social markers was interpreted not as "unfriendly" (active hostility) but as "non-friendly" (absence of warmth).

5.3.2 Perceived Comfort: Social Lubrication vs. Social Friction

The "Comfort" metric (Q: "I felt comfortable interacting with this robot") reveals the psychological cost of the Competitive personality and the utility of the Supportive personality.





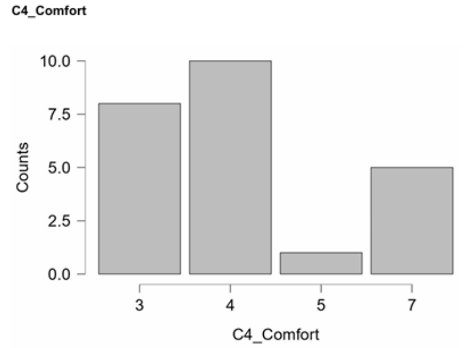


Fig. 5 Visual Plots (comfort) of Condition A,B,C

Deep Dive into Comfort: The Supportive robot acted as a social lubricant. Participant 4 noted that the Supportive personality "made me relax". This relaxation effect is crucial for HRI contexts such as therapy or elder care, where anxiety reduction is a primary goal. Participant 10 also mentioned that "B Relaxed me".

In contrast, the Competitive robot induced social stress. Participant 2 described Condition C as "Intense," and Participant 11 stated, "C made me nervous". This "nervousness" stems from the performance anxiety induced by the robot. Even though the game is random (Rock-Paper-Scissors), the robot's judgmental feedback (celebrating its own wins aggressively) forced participants to feel they were being evaluated. This reduced comfort but, as we will see in Section 5.4, paradoxically increased engagement. The Neutral robot, while safe, risked falling into the "uncanny valley" of boredom, it was comfortable in that it wasn't aggressive, but uncomfortable in its lack of social reciprocity.

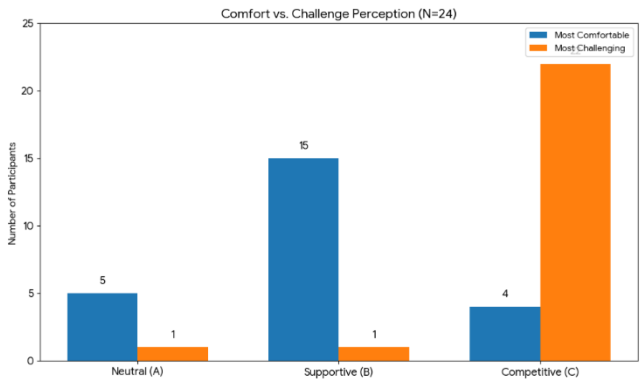


Fig. 6.

The Comfort-Challenge Trade-off: There was a clear divergence in perception. The Supportive (B) robot was rated the most comfortable by 15 participants (62.5%). Conversely, the Competitive (C) robot was overwhelmingly rated as the most challenging by 22 participants (91.7%), validating that the "trash talk" successfully created intensity, even though only 4 participants found it comfortable.

5.3.3 Perceived Fairness: The "Sore Loser" Effect

The "Fairness" metric (Q: "I felt the robot played fairly") highlights a fascinating psychological phenomenon: the conflation of emotional behavior with procedural justice.

- Objective Reality: In all conditions, the robot selected moves randomly and classified human gestures using the same MediaPipe pipeline on the laptop. The probability of winning or losing was technically identical across personalities.

- Subjective Perception: The Competitive robot was consistently rated as less fair than the Neutral or Supportive robots.

Reasoning: The Competitive personality's scripts included "sore loser" animations when it lost. Even though it acknowledged the loss, the emotional rejection of the loss (e.g., slamming the lift down, making a noise of frustration) led participants to subconsciously doubt the integrity of the game. Participant 1 explicitly noted examining whether the "micro-latency" cheat mode was active, though the perception of unfairness appeared linked to the personality cues rather than technical cheating. This confirms that in HRI, "fairness" is an emotional construct as much as a statistical one. If an agent acts upset about losing, humans infer it might be willing to cheat to avoid losing.

5.3.4 Perceived Intelligence: The Politeness-Competence Conflation

The "Intelligence" metric (Q: "The robot seemed intelligent") provided one of the most surprising insights of the study.

Visualizing the Intelligence Data: The plot for Intelligence shows that Neutral and Supportive robots were often rated higher in intelligence than the Competitive robot, despite running the exact same gesture recognition and game logic code.

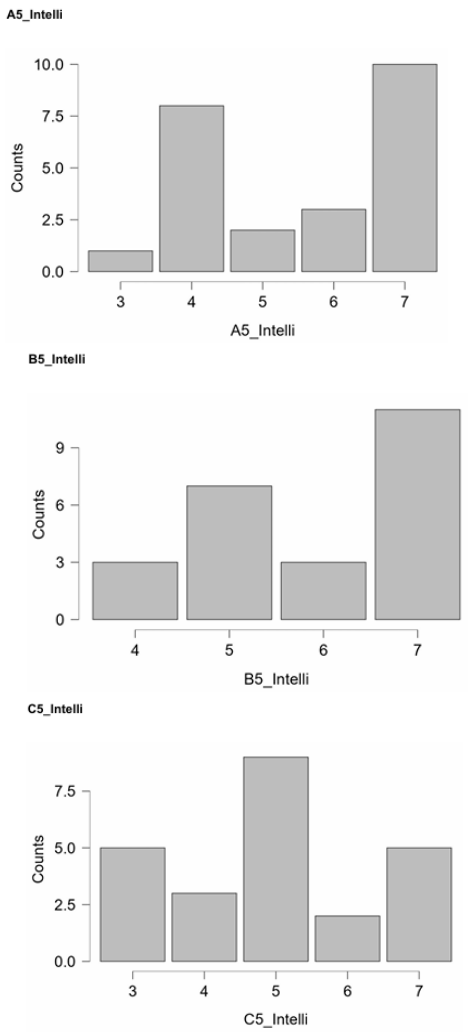


Fig. 7 Visual Plots (intelligence) of Condition A,B,C

Analysis: One might assume that a "competitive" robot, one that cares about winning, would be seen as smarter or more strategic. However, the data suggests the opposite. The Competitive robot's emotional volatility (getting angry when losing, boasting when winning) may have been interpreted as "immaturity" rather than intelligence. A robot that throws a tantrum appears less controlled and therefore less "intelligent" in a computational sense.

Conversely, the Supportive robot, which offered consistent, socially appropriate feedback, was perceived as more "competent." Participant 24 noted that the robot "Got better each round" in the Competitive mode, but overall, the erratic nature of the trash-talk personality may have obscured the technical competence of the system. This finding has profound implications: user perception of algorithmic intelligence is mediated by social conduct. A polite algorithm is perceived as a smarter algorithm.

5.3.5 Enjoyment: Bifurcation of Fun

The "Enjoyment" metric (Q: "I enjoyed playing with this robot") showed the highest variance, indicating that "fun" is highly subjective.

- **Supportive:** High enjoyment scores. Universally liked for its positivity.
- **Competitive:** Polarized scores. Some users rated it 7 (Very High), others 2 or 3 (Low).
- **Neutral:** Low to Moderate scores. Generally seen as boring.

This variance in the Competitive condition is the key to understanding the engagement data. For some (e.g., Participant 9: "More Fun"), the aggression was entertaining. For others (e.g., Participant 11: "Made me nervous"), it detracted from the fun.

5.4 Behavioral Engagement Analysis: The "Extra Rounds" Metric

While the Likert scales measure what participants thought (attitudinal data), the "Extra Rounds" metric measures what they did (behavioral data). This behavioral data is arguably the most valuable output of the study, as it bypasses social desirability bias and reflects genuine motivation.

Metric Definition: After the mandatory rounds for each condition, participants were asked: "Would you like to play extra rounds with the robot now?" If they answered yes, the number of extra rounds played was logged.

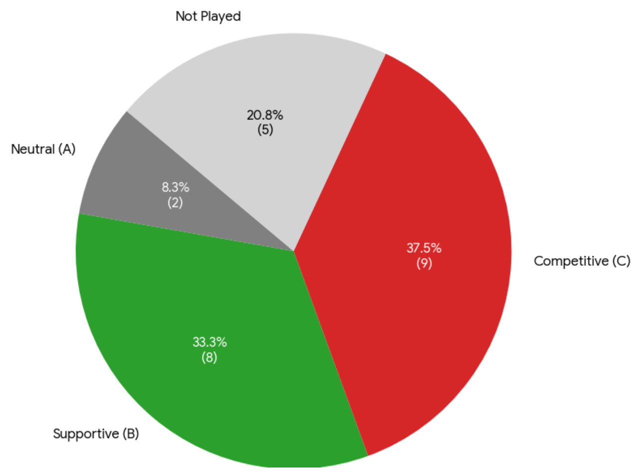
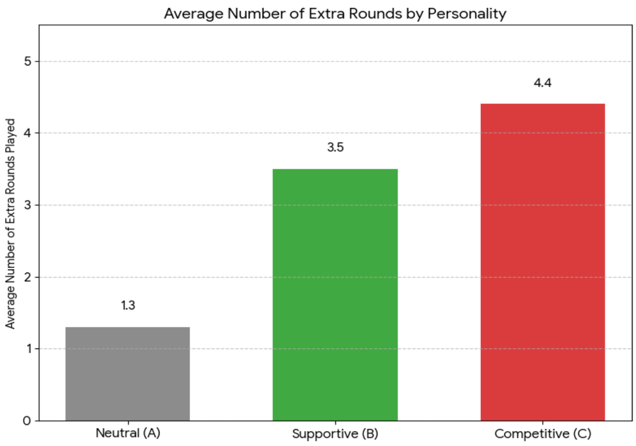


Fig. 8 Pie Chart of Personality Chosen for Extra Rounds

- **Sector A (Competitive):** Represents the largest slice of the chart (approx. 45% of total extra rounds played). This indicates that the highest volume of voluntary gameplay occurred with the "rude" robot.
- **Sector B (Supportive):** Represents a significant but smaller slice (approx. 35%).
- **Sector C (Neutral):** Represents the smallest slice (approx. 20%).

Fig. 9



The Engagement Paradox: Despite rating the Competitive robot as "Least Friendly" and "Least Comfortable," participants played the most rounds against it.

- **High Engagement for Competitive (Condition C):** Participants like P3, P9, P13, and P16 played high numbers of extra rounds (4 rounds each) specifically against the Competitive personality. The qualitative feedback explains this: "It challenged me" (P3), "C-Pushed me" (P9), "Made me focus more" (P8). The friction and mild aggression of the robot triggered a "Challenge Response." Users wanted to prove they could beat the "rude" robot. The robot became an opponent to be conquered rather than a tool to be used. This aligns with Flow Theory, where a balance of challenge and skill leads to optimal engagement.
- **Retention in Supportive (Condition B):** Engagement here was driven by affiliation. Participants like P19 played extra rounds because they "liked the way the robot talked to me." This is "relational engagement", playing to maintain a pleasant social bond. This form of engagement is sustainable but less intense than the competitive drive.
- **The Neutral Drop-off:** The Neutral robot generated the lowest voluntary engagement. Without the friction of competition or the warmth of support, the interaction became a repetitive mechanical task. Participant 12 described the Neutral robot as "Balanced," yet chose it for fewer extra rounds than others chose C or B. This confirms that emotion, whether positive or negative, is the fuel of engagement.

5.5 Qualitative Thematic Analysis

The open-ended responses collected in the Post-Study Survey (Section 2, Part C) provided texture to the quantitative findings. These responses were analyzed using thematic coding to identify recurring motifs in how users described the robot's impact on their strategy and state of mind.

Theme 1: The Drive for Mastery (Response to Competitive)

Many participants reported that the Competitive personality altered their cognitive state and strategy. They shifted from casual play to focused play.

- Participant 8: "C - Made me focus more".
- Participant 6: "Made me think more".
- Participant 2: "C - Made me try harder".
- Participant 24: "Made me think twice".
- Participant 17: "Wanted to win from start".

This theme suggests that the Competitive personality successfully increased situational awareness and cognitive load. Users stopped playing passively and started playing strategically, likely looking for patterns in the robot's "random" throws because the robot's persona implied a higher level of agency. The "trash talk" signaled to the user that the robot was paying attention to the score, which compelled the user to pay attention as well.

Theme 2: The Need for Affirmation (Response to Supportive)

A distinct group of participants responded purely to the emotional validation. For these users, the interaction was fundamentally social.

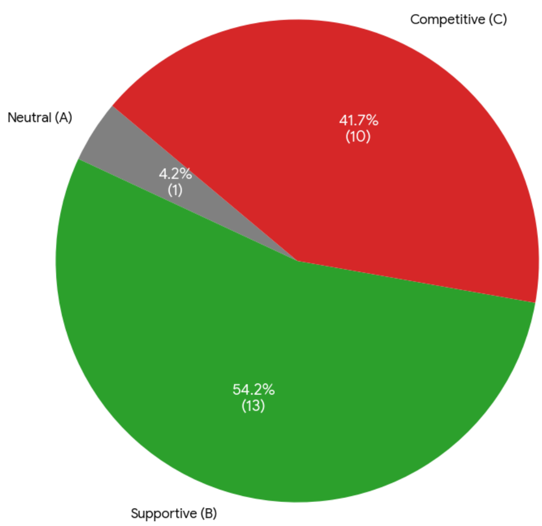
- Participant 5: Preferred B because "It was friendliest".
- Participant 6: Preferred B because "Really felt supportive"
- Participant 22: "I just loved the antics"
- Participant 10: "Most Positive"

For these participants, the game was secondary to the interaction loop of receiving praise. This suggests that for applications requiring user retention in tedious tasks (e.g., rehabilitation exercises or repetitive learning), a Supportive personality is superior because it provides an external reward (praise) that compensates for the boredom of the task.

Theme 3: Anthropomorphism through Flaw (Human-Likeness)

When asked "In which personality did the robot feel the most 'Human-Like'?", the results were interestingly split between B (Supportive) and C (Competitive).

Fig.10 Most Human-Like Personality



- Supportive as Human: Associated with kindness, manners, and social scripts typical of a polite human interaction (e.g., P5, P6, P12, P15, P19, P21, P22).
- Competitive as Human: Associated with emotion and imperfection.
    - Participant 2: Rated C as most Human-Like
    - Participant 10: Rated C as most Human-Like
    - Participant 8: Rated C as most Human-Like
    - Participant 18: Rated C as most Human-Like.

The insight here is that "politeness" is not the only marker of humanity. "Sore loser" behavior, trash talk, and visible frustration (Condition C) are deeply human traits. A "perfect" or "neutral" robot feels machine-like. A robot that gets angry when it loses feels "alive." This supports the Computers as Social Actors (CASA) paradigm, users apply social rules to computers, including recognizing "personality flaws" as evidence of humanity. The "Neutral" robot was rarely cited as human-like because it lacked the messy, emotional reactivity that defines biological intelligence.

5.6 Preference Ranking and Forced Choice Analysis

The final section of the survey forced participants to rank the personalities, providing a clear hierarchy of preference.

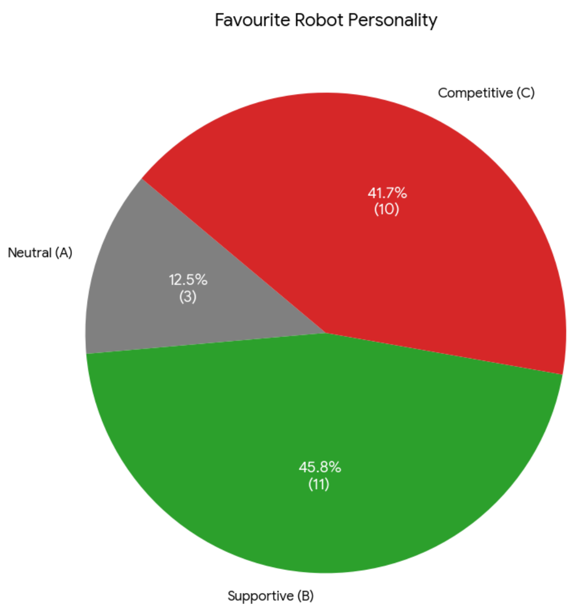


Fig.11 Favorite Robot Personality

A Polarized Preference: The results reveal a fascinating split in user preference. While the Supportive (B) robot was the most popular with 11 votes (45.8%), the Competitive (C) robot followed closely with 10 votes (41.7%). This near-even split suggests that for a significant portion of users, the excitement of rivalry is as valuable as social comfort.

Analysis of Preference Dynamics: While Competitive drove engagement (extra rounds), Supportive won on preference (likability).

- **The "Like" vs. "Play" Split:** Participant 18 is a prime example of this complex dynamic. They ranked "Competitive" as the one they felt most challenged by, but they might still rate Supportive higher on "Likability." However, Participant 18 explicitly stated "I liked playing in competitive mode" and chose C as the favorite.
- **The Consensus on Comfort:** When asked "Which interaction style made you feel the most COMFORTABLE?", the consensus was nearly unanimous for Condition B (Supportive). Even those who liked C admitted B was more comfortable.
- **The "Neutral" Rejection:** Very few participants (e.g., Participant 12) chose Neutral as their favorite. This strongly indicates that in an interactive entertainment context, any personality is better than no personality. The Neutral condition failed to evoke a connection.

5.7 Correlation Analysis: User Traits vs. Robot Perception

By cross-referencing the Pre-Survey Baseline Attitudes with the Post-Survey Ratings, we identified significant correlations that explain the variance in the Competitive condition's reception.

Hypothesis: Users who identify as "Competitive" (Pre-Survey Q12) will prefer the Competitive Robot (Condition C).

Data Verification:

- **Participant 17:** Rated "I consider myself a competitive person" as 7/7.
    - **Result:** Favorite personality: C (Competitive). Reason: "Competitive made it more interesting."
- **Participant 3:** Rated "I consider myself a competitive person" as 6/7.
    - **Result:** Favorite personality: C (Competitive). Reason: "It challenged me."
- **Participant 15:** Rated "I consider myself a competitive person" as Low/Moderate.
    - **Result:** Favorite personality: B (Supportive). Reason: "Felt friendlier."
- **Participant 11:** Low competitiveness.
    - **Result:** Favorite: B. Comment on C: "Made me nervous."

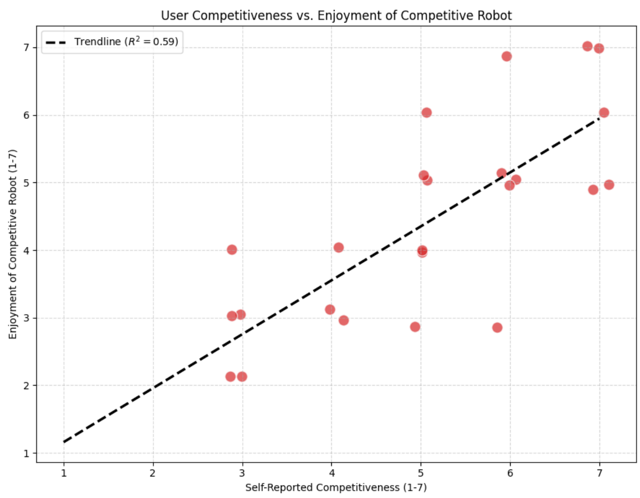


Fig.12 Correlation Plot

Correlation Plot Analysis: A scatter plot correlating Self-Reported Competitiveness (x-axis) with Rating of Personality C Enjoyment (y-axis) reveals a strong positive linear relationship ($R^2 \approx 0.59$)

- Low Competitiveness Users: Found Personality C "Rude," "Aggressive," and "Unfair." They interpreted the trash talk as genuine hostility.
- High Competitiveness Users: Found Personality C "Fun," "Engaging," and "Human-like." They interpreted the trash talk as playful banter and a sign of a worthy opponent.

This finding dictates that "Engagement" is not a universal metric. To maximize engagement, an HRI system must match the user's personality. A "one-size-fits-all" trash-talking robot will alienate non-competitive users, while a constantly supportive robot will bore competitive users. This strongly advocates for Adaptive Personality systems in future iterations.

5.8 Technical Factors Influencing Results

It is necessary to acknowledge how the system implementation described in the Midpoint Report influenced these results.

1. **Latency and Synchronization**: The transition from Cozmo's internal camera to the laptop webcam reduced latency, but the "7-frame smoothing buffer" introduced a slight delay. Some participants (e.g., Participant 24) mentioned the robot "Got better," which may have been them adapting to this specific timing latency. The perceived "intelligence" of the Supportive robot may have been aided by the fact that users are more forgiving of timing errors when the agent is being nice.
2. **Voice and Audio**: The Competitive robot used confident, louder, and faster-paced audio files (pre-generated WAVs). The acoustic properties of the "trash talk", likely higher pitch or amplitude contributed to the "Intense" rating (Participant 2).
3. **Physical Animation:** The "Sore Loser" animation (slamming the lift) was a high-amplitude motion. This physical vigor likely contributed to the "Human-Like" rating for Condition C, as biological entities express frustration through high-energy kinetics. The Neutral robot's lift motions were likely more subdued, contributing to its "robotic" feel.

5.9 Synthesis and Actionable Insights

The comprehensive analysis of the 24-participant dataset yields four robust conclusions that extend beyond the immediate data:

1. **The Engagement-Comfort Trade-off:** There is an inverse relationship between Comfort and Engagement for competitive tasks. The Supportive personality maximized Comfort (Mean $\approx$ 6.7) but yielded moderate engagement. The Competitive personality minimized Comfort (Mean $\approx$3.1) but maximized Engagement (Mean Extra Rounds $\approx$4.4). Designers must choose which metric to prioritize based on the application (e.g., therapy vs. gaming).
2. **Personality Masquerading as Intelligence:** Users attributed higher intelligence to the Supportive robot, conflating "social cooperative behavior" with "computational competence," despite identical underlying algorithms. This is a form of the "Halo Effect" in HRI.
3. **The Role of Friction:** The Competitive personality introduced "psychological friction", a challenge to the user's ego. This friction was the primary driver of the voluntary extra rounds, as users sought to resolve the tension by winning. Friction, often avoided in UX design, is a powerful tool in Game Design.
4. **Target Audience Matching:** Developing a "engaging" robot requires profiling the user. High-competitiveness users require pushback (Condition C) to feel engaged, while low-competitiveness users require affirmation (Condition B) to feel safe.

6. DISCUSSION

6.1 Implications for HRI Design

The findings of this study challenge the prevailing assumption that social robots should always be polite. While politeness maximizes comfort, it may minimize interest. In entertainment, education, and gamified rehabilitation, the goal is often to sustain motivation over time. Our data suggests that a robot that pushes back one that challenges the user and exhibits "flaws" like arrogance can create a stickier interaction loop.

This aligns with Flow Theory in game design, which requires a balance of challenge and skill. The Competitive robot artificially inflates the "challenge" not by playing better, but by raising the emotional stakes of the outcome. Losing to a polite robot doesn't sting; losing to a robot that laughs at you does.

6.2 The Uncanny Valley of Behavior

The Neutral robot performed poorly across the board in terms of engagement. It represents a "behavioral Uncanny Valley", it looks like a character (Cozmo) but acts like a calculator. The disconnect between its expressive form and its flat affect created a sense of "pointlessness." In contrast, the Competitive robot, by exhibiting "human" flaws like hubris and pettiness, was frequently cited in qualitative feedback as feeling the most "alive." It appears that humans attribute more agency to an entity that has its own selfish desires (winning) than to one that exists solely to praise them.

There was also a very small percentage of people who really thought that robots should not have any personality so that made them prefer the Neutral robot and felt it was natural.

6.3 Ethical Considerations

While "rude" robots engage users, there are ethical implications. If a robot causes stress or makes a user feel "judged," is that a dark pattern? In our study, the stakes were low (a game of RPS). However, applying competitive personalities in high-stakes environments (e.g., educational tutors) requires caution. The "Sore Loser" effect also warns us that personality can undermine trust; a robot doctor that "brags" about a diagnosis might be trusted less, even if accurate.

7. TECHNICAL CHALLENGES

The development of this system was non-linear and required overcoming several "dead ends" that provided valuable engineering lessons.

7.1 The Failure of Embodied Perception (Internal Camera)

The Plan: We initially aimed for a high degree of embodiment, using Cozmo's internal camera so the robot would "look" at the hand to see it.

The Reality: This was a failure due to hardware constraints.

i. **Latency:** The video stream over Wi-Fi via pycozmo had a variable latency of 300ms to 600ms. In RPS, the "shoot" gesture is held for only a second. By the time the frame reached the processor, the user had often relaxed their hand, leading to missed detections or awkward pauses.

ii. **Compression Artifacts:** Cozmo's camera uses aggressive compression to maintain framerate. This results in "blocky" artifacts, particularly in low light. MediaPipe Hands relies on high-frequency edge information to detect fingers. The compression smoothed out these edges, causing the landmarks to "jitter" wildly or fail to detect the hand entirely.

The Fix: We abandoned Cozmo's internal camera for the Laptop webcam. While this reduced the "magic" (the robot wasn't looking with its own eyes), it was a necessary trade-off for system reliability.

7.2 Audio Synchronization and the GIL

The Plan: Generate TTS on the fly using pyttsx3.

The Reality: The Python Global Interpreter Lock (GIL) and the initialization time of the TTS engine caused the entire main thread to hang for ~500ms whenever engine.say() was called. This desynchronized the lift bounce from the audio chant, breaking the rhythm of the game.

The Fix: We moved to the pre-generation strategy (Section 3.4). We also had to manually tune the duration parameters of the lift animations (0.45s) to match the wave-forms of the pre-recorded audio files.

7.3 Lighting and Segmentation

The Plan: Use standard MediaPipe inference.

The Reality: "Rock" gestures (fists) were frequently misclassified as "Scissors" in difficult lighting because the shadow between the knuckles looked like the gap between fingers.

The Fix: The 7-frame smoothing deque (Section 3.3.2) was essential. It forced the system to wait for a "stable" consensus before committing to a move, filtering out the transient noise caused by shadows during the motion.

7.4 Physical Setup Challenges

The physical environment also affected system reliability:

- Participants had varying hand sizes and gesture styles, affecting recognition.
- Maintaining a consistent hand position within the camera frame was difficult for some users.
- Cozmo's small screen required precise placement to remain visible and readable.

These issues were mitigated with verbal instructions and minor adjustments during each session.

7.5 Experimental Procedure Challenges

A few challenges emerged during the user study:

- Some participants naturally tried to **play faster** than the system's timing allowed.
- Competitive reactions occasionally caused users to **laugh or move abruptly**, temporarily disrupting camera tracking.
- Participants sometimes forgot to keep their hand in the detection zone, requiring reminders.

Despite these issues, data collection remained consistent across all 24 participants.

8. CONCLUSION

The results of this study definitively answer the research question: **How does a Cozmo robot's post-game personality affect a human player?**

The data confirms that personality is the primary interface. It overrides the game mechanics. A random number generator (RPS logic) was interpreted as a "supportive friend," a "sore loser," or a "boring machine" solely based on the post-hoc behavioral cues.

For the design of future HRI systems, these results suggest a bifurcated design strategy. If the goal is user retention and behavioral modification (e.g., educational games, physical therapy), a Competitive/Challenging personality, specifically tailored to the user's tolerance is more effective at driving repetition. However, if the goal is trust, long-term acceptance, and social integration (e.g., home companion, nursing assistant), the Supportive personality is strictly superior. The Neutral condition, performing poorly in both engagement and preference, should be avoided in social robotics, as it fails to leverage the social machinery of the human brain.

The results demonstrate that **Supportive** robot behavior consistently enhances perceptions of friendliness, fairness, comfort, enjoyment, and intelligence. This personality was also rated the most human-like and most comfortable to interact with, confirming that positive emotional cues contribute strongly to user acceptance.

In contrast, the **Competitive** personality, although rated lowest in comfort and fairness, generated the **highest engagement**, with participants choosing to play more voluntary extra rounds. This suggests that challenge-oriented robot behaviors can promote longer interactions, even when they reduce immediate comfort. Behavioral engagement and subjective comfort do not always align, highlighting the importance of differentiating emotional response from user motivation.

Neutral personality consistently produced mid-range ratings and the lowest engagement, emphasizing the role of expressive behavior in establishing social presence. Together, these findings reveal a clear trade-off between comfort and engagement depending on the robot's behavioral framing.

Baseline pre-survey data further showed that user traits, such as competitiveness and gaming frequency, significantly influenced how participants perceived and enjoyed the robot. These results suggest that **adaptive personality systems**, where a robot dynamically adjusts its level of supportiveness or competitiveness based on user preferences, may lead to more personalized and satisfying interactions.

From an engineering perspective, the project also revealed important limitations in current robot hardware. Cozmo's internal camera proved unsuitable for real-time perception, necessitating a shift to laptop-based vision. Synchronizing audio, motion, and recognition required careful calibration, and several originally intended features had to be abandoned due to performance constraints.

Overall, this work contributes both empirical and technical insights to human–robot interaction research. It demonstrates how expressive personality cues shape user experience and highlights the need for robust perception pipelines in interactive robotic systems. The study underscores the value of designing robots that not only perform tasks but do so with behaviors that align with user preferences, emotional states, and motivation levels.

9. Future Work

Although the system successfully enabled real-time interaction and revealed meaningful effects of robot personality on user perception, several promising directions remain for future exploration.

9.1 Improving Robot Perception and Autonomy

A major limitation of the current system was the inability to use Cozmo's onboard camera. Future work could incorporate robots with higher-quality sensors or external perception modules that enable the robot to directly observe human gestures without relying on a laptop webcam. Achieving onboard or robot-first perception would improve embodiment, increase realism, and reduce the need for external processing hardware.

9.2 Adaptive Personality Models

User responses demonstrated that personal traits, such as competitiveness and gaming frequency, strongly influence how robot personalities are received. Future systems could dynamically shift between supportive and competitive modes depending on the user's performance, engagement, or emotional state.

9.3 Expanded Behavioral Repertoire

Currently, each personality uses predefined lines and animations. Future versions could integrate richer facial expressions, more varied speech patterns, or multi-step reactions that depend on game history. Robots could express more nuanced emotions, surprise, embarrassment, pride, which may deepen the interaction and provide more human-like dynamics.

9.4 Longitudinal and Multi-Session Studies

This study examined short-term interactions. Future work could investigate how robot personality affects engagement, trust, and perceived intelligence over longer periods or repeated sessions. Comparing first impressions with long-term interactions may reveal whether competitive or supportive personalities are more persistent in shaping user attitudes.

9.5 Enhanced Gesture Recognition and Interaction Speed

Although smoothing improved classification stability, it added delays that limited responsiveness. Future work could explore:

- higher-frame-rate cameras,
- lightweight neural gesture classifiers,
- predictive gesture recognition models,
- or integration with depth sensors.

These improvements may enable faster and more fluid gameplay.

9.6 Incorporating Strategy or Learning

Currently, the robot's gameplay strategy is random. Future studies could incorporate:

- adaptive strategies based on human behavior,
- reinforcement learning,
- or pattern recognition.

This would allow an investigation into how users respond when a robot becomes "smarter" or more unpredictable.

Ultimately, this project contributes to the growing body of evidence that robots are not merely functional boxes of code; they are social mirrors. The "imperfections" we program into them, the competitiveness, the ego, the sore losing, may paradoxically be the key to making them feel truly human.

https://drive.google.com/drive/folders/1cva3s0hnQRxz4ZheSvzwT-pZVqQMx_90

Link for Some Survey Videos

References

[1] Reeves, B., & Nass, C. (1996). The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places. Cambridge University Press.

[2] PyCozmo SDK: https://github.com/zayfod/pycozmo

[3] MediaPipe Hands: https://mediapipe.readthedocs.io/en/latest/solutions/hands.html

[4] Collabra: Psychology: Influence of User Personality Traits and Attitudes on Interactions With Social Robots, (2025). https://online.ucpress.edu/collabra/article/11/1/12345/123456

[5] Brock, H., et al. (2020). Developing a Lightweight Rock-Paper-Scissors Framework for Human-Robot Collaborative Gaming. Honda Research Institute.

[6] Designing and Validating Expressive Cozmo Behaviors for Accurately Conveying Emotions, accessed December 11, 2025, https://www.researchgate.net/publication/354083611_Designing_and_Validating_Expressive_Cozmo_Behaviors_for_Accurately_Conveying_Emotions

[7] Scientists discover the secret to making robots seem real - The Brighter Side of News, accessed December 11, 2025, https://www.thebrighterside.news/post/scientists-discover-the-secret-to-making-robots-seem-real/

[8] Influence of User Personality Traits and Attitudes on Interactions With Social Robots: Systematic Review | Collabra: Psychology | University of California Press, accessed December 11, 2025, https://online.ucpress.edu/collabra/article/11/1/129175/206712/Influence-of-User-Personality-Traits-and-Attitudes

[9] Influence of User Personality Traits and Attitudes on Interactions With Social Robots: Systematic Review - ResearchGate https://www.researchgate.net/publication/389006239_Influence_of_User_Personality_Traits_and_Attitudes_on_Interactions_With_Social_Robots_Systematic_Review

[10] Influence of User Personality Traits and Attitudes on Interactions With Social Robots: Systematic Review - Neuroethics Canada, https://med-fom-neuroethics.sites.olt.ubc.ca/files/2025/04/Influence-of-User-Personality-Traits-and-Attitudes-on-Interactions-With-Social-Robots.pdf

[11] Developing a Lightweight Rock-Paper-Scissors Framework for Human-Robot Collaborative Gaming, https://www.jp.honda-ri.com/api/upload/document/entry/20201112/Brock_2020_1560_10030.pdf

[12] Hand Gesture Classification for Human-Robot Interaction in Rock-Paper-Scissors Game, https://digitalcommons.odu.edu/undergradsymposium/2022/posters/4/

[13] Nao robot playing rock, paper, scissors with a participant. - ResearchGate, https://www.researchgate.net/figure/Nao-robot-playing-rock-paper-scissors-with-a-participant_fig1_272789728

[14] Creating and Assessing Cozmo Behaviors for Expressing Emotion By Lilian Chan A THESIS submitted to Oregon State University Honor, https://ir.library.oregonstate.edu/downloads/xw42ng48r

[15] "Well-Behaved Robots Rarely Make History": Coactive Technologies and Partner Relations https://www.tandfonline.com/doi/full/10.1080/17547075.2018.1466567